IMPLEMENTATION PLAN V2 · FLOLABS RECRUITING TEAM · 31 OCT 2026

Manager Buddy

An AI teammate added next to our CRM. You talk to it by chat or voice. It acts with your permissions, remembers everything, works on its own, finds and follows up leads, warns us when something breaks, and shows what it costs, live.

TWO RULES FOR EVERY STEP OF THIS PLAN

1 **Nothing in the existing CRM is removed or changed**

Buddy is added next to the CRM, not inside it. It uses the CRM's existing features through the APIs the chatbot already uses. Existing code, screens, workflows and data model stay exactly as they are.

2 **Company data is protected, and every change is tested before and after**

Buddy never touches the CRM's database directly. It reads and writes only through the CRM, with the permissions of the person it acts for. Every release is tested on a copy of production data first, and the full CRM is re-tested afterwards.

Changes to existing CRM code	Time to full scope	Expected run cost	Cost visibility
0	**12 weeks**	**~$280**	**Live**
in every phase of this plan	useful from week 3	a month with the optimizer	token meter from phase 1

INSIDE THIS PLAN

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Replaces the "Manager Buddy" proposal of 28 Oct and the implementation plan of 30 Oct. Facts about our CRM were checked in the code on the automations_branch. Prices for Claude were checked on Anthropic's pricing page on 31 Oct 2026; other vendor prices come from third-party sources and are marked as estimates.

Your brief, and six things Buddy will do

In your words

> "It shouldn't interfere with the CRM's existing functionality at all."
> ON THE CRM

> "We can't change the base CRM. We have company data there."
> ON THE CRM

> "What the agent can do will depend on the access the user has."
> ON ACCESS

> "It needs to remember everything."
> ON MEMORY

> "Assign it tasks and schedule them. It should work autonomously: replying to email, sending us alerts, telling us if a workflow has failed."
> ON AUTONOMY

> "It would be chat, but more of a voice agent."
> ON VOICE

> "Keep us on new leads, keep in contact with old ones, and if someone is interested, inform us. Daily reports on everything."
> ON LEADS

> "We should see our token usage live."
> ON COST

Six capabilities

Chat and voice

Talk to Buddy in the browser, on your phone or by email. Same brain behind each.

Acts as you

It can only do what your CRM role allows. A read-only user gets a read-only Buddy.

Remembers everything

Conversations, emails, people, leads and decisions, searchable by meaning and over time.

Works on its own

Tasks, schedules, email replies, alerts, monitoring and daily reports.

Finds and follows leads

Apify search, outreach, nurture, alerts when someone is interested. Drafts workflows.

Cost under control

Estimator, optimizer, budgets and a live token meter.

How we keep the two rules, in practice

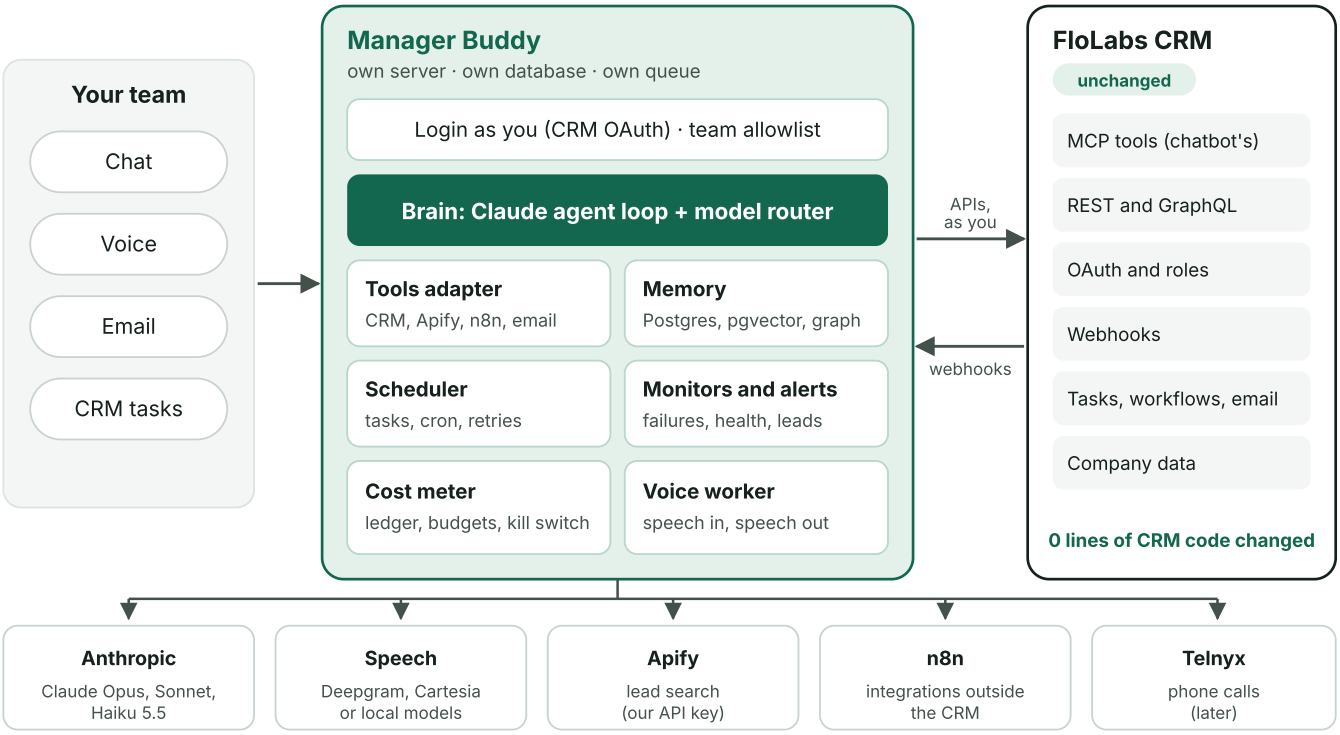
Safeguard	What it means
Separate service	Buddy runs on its own server with its own database, queue and logs. If Buddy stops, the CRM does not notice.
Existing APIs only	MCP (the chatbot's tools), REST and GraphQL, OAuth and webhooks. No direct database or Redis access, ever.
The CRM decides permissions	Buddy acts with the user's own role. Buddy's app role adds a ceiling on top. Both must allow an action.
Everything is traceable	Every record Buddy creates or edits is logged with who asked and why, so it can be listed and reverted.
Test before and after	Every release: CRM regression suite on a staging copy of production, Buddy on, suite again, then production right after a backup.
CRM touchpoints register	If the CRM ever needs a change, it is new files only, switched off by default, tested on staging and listed here. Today the register is **empty**.

CRM-SAFE This green note appears in every section. It says exactly how that part of Buddy stays outside the CRM.

02 · ARCHITECTURE

Next to the CRM, connected only through its APIs

One service with clear parts. The only part that knows it is talking to our CRM is the tools adapter, so a CRM upgrade can only break Buddy in one place, and never the other way around.



What we reuse from the CRM, as it is

CRM feature (exists today)	How Buddy uses it	Checked in our code
MCP endpoint	The same tools the chatbot uses: find, search, group by, create notes and tasks, email, workflows. Buddy shows Claude only the ones we pick.	`engine/api/mcp`
OAuth server	Each person connects Buddy once. Buddy then acts as them, with refresh tokens.	`application-oauth` , self-registration
Roles and permissions	Checked by the CRM on every call Buddy makes.	`resolveCallerRoles`
Webhooks	New email, form submitted, record changed, workflow run finished.	webhook jobs
Tasks and workflows	Assign a task to the Buddy user. List failed runs, draft new workflows.	`list_workflow_runs` and 14 more tools
AI usage records	The chatbot's token use, shown on the same cost dashboard.	`agentTurn` (tokens, credits)
Apps with front components	Optional voice button and cost widget inside the CRM, with a "Use your microphone" permission.	front-component renderer

CRM-SAFE Everything above is used through its public interface, the way the chatbot or an outside app would use it. No CRM file is edited.

03 · ACCESS

Buddy can do what you can do, and nothing more

You said role-based access is already in the CRM, and you are right. We checked how it treats an outside app that acts for a person: the CRM allows an action only if **both** the person's role and the app's own role allow it. So Buddy inherits the access model without a line of new permission code.

You ask
"Move these 12 people to Contacted."

→

Buddy calls the CRM
with your login token, through the chatbot's tools.

→

The CRM checks
your role, and Buddy's app role. Both must allow it.

→

Allowed: done and logged
Not allowed: refused, and Buddy tells you why.

Checked in code: `resolveCallerRoles` in `mcp-protocol.service.ts` and `build-user-role-ids-by-source.util.ts` ("An application acting for a user is bounded by both the user's role and its own"). The CRM's OAuth server accepts self-registered apps with authorization-code and refresh-token grants (`oauth-registration.controller.ts`). End-to-end login is tested on staging in phase 0.

Who Buddy acts as

Situation	Buddy acts as	Result
You chat or talk to Buddy	You	A read-only user gets answers but no edits. Someone with write or delete access can have Buddy write or delete, after a confirmation.
A task you assigned or scheduled	You (saved login)	If your access changes or you leave, the task stops and tells the team. It never runs with more than you have today.
Team jobs: leads, monitoring, daily reports	The Buddy user	A normal CRM user with its own role, set by an admin.
Always, on top	Buddy's app role	A ceiling for everyone, admins included. We suggest it excludes settings, roles and data-model changes.

Two ways to reach the chatbot's functions

	A. Existing APIs **Recommended**	B. Special interface inside the CRM
CRM code changed	**None**	New module (added files) plus one registration line
Acts as the user	Yes, through the CRM's OAuth	Yes, in the CRM's own process
Which functions	Same tools as the chatbot via `/mcp` . Buddy exposes only the ones we choose.	Same tool registry, custom subset
If Buddy fails	CRM unaffected, different server	Runs inside the CRM, so a bug can reach it
CRM upgrades	Contract tests catch API changes	Must be re-tested with every upgrade
When	From day one	Only for a gap the APIs cannot cover

CRM-SAFE We start with A. If we ever need B, it is built as new files behind a switch that is off by default, tested on a copy of production, and recorded in the CRM touchpoints register. API load is capped by the CRM's own rate limits for apps, and Buddy paces itself below them.

How Buddy remembers everything

Remembering everything is mostly a storage and search problem, not an AI problem. We keep a complete record in Buddy's own database, in layers, and memory is never allowed to become a way around permissions.

1. Facts	Stages, counts, who replied, who signed. Never copied as truth. Always read live from the CRM, with your permissions.	`CRM, live`
2. Event log	Every conversation, voice transcript, email handled, action, alert and run. Append-only, searchable by person and date.	`Buddy Postgres`
3. Search by meaning	Emails, notes, form answers, transcripts and lead research, as vectors, plus keyword search for names and emails.	`pgvector + full-text`
4. Knowledge graph	People, companies, leads, team members and campaigns, joined by dated links: replied to, interested in, met, referred by, works at.	`Postgres graph tables`
5. Team rules	"Never email applicants on Fridays." Small, edited by the team, loaded every time. Plus Buddy's own notes through Claude's memory tool.	`Buddy Postgres`
6. Working memory	The conversation in hand. Long threads are summarised automatically (compaction) and repeated context is cached.	`Claude context`

Options we looked at

Option	What it is	License	Verdict
Postgres + pgvector + full-text	Vectors and keywords in one database we already know how to run and back up	PostgreSQL	Start here
Apache AGE	Graph queries (openCypher) inside Postgres. Current releases target Postgres 17 and 18	Apache-2.0	If needed
Graphiti (with Neo4j)	Temporal knowledge graph: old facts expire instead of being overwritten. Pre-1.0, AI cost on every write	Apache-2.0 (Neo4j CE: GPLv3)	Trial later
Mem0	Memory layer for preferences and facts, vector and graph	Apache-2.0	Alternative
Letta, Cognee	Full agent runtime with memory tiers (Letta); graph and vector knowledge engine (Cognee)	Apache-2.0	Alternative
Kùzu	Embedded graph database	MIT, archived Oct 2025	Avoid
FalkorDB	Graph database used by some memory tools	SSPLv1, source-available	Avoid
Anthropic memory stores	Memory hosted by Anthropic (Managed Agents, beta)	Service	Not now

How much storage

43k people and about 100k emails is roughly 150M tokens of text. As vectors plus text and indexes that is about 2 to 4 GB, growing 1 to 2 GB a year. Transcripts only, no audio, unless we decide otherwise.

Permissions stay intact

Each memory points to its source record. Before Buddy shows a memory to you, it re-checks that record with your login. If you can't open it in the CRM, you don't see it through Buddy.

Forgetting on request

When a person is deleted in the CRM, a webhook tells Buddy to delete their memories too (GDPR). Embeddings come from a local model by default, so text does not leave our servers.

CRM-SAFE Memory lives in Buddy's own Postgres on Buddy's server. It is filled from webhooks and from slow, throttled reads at night. It never connects to the CRM's database and never writes back to it.

Give it work, let it run, hear about what matters

Say it
Chat or voice: "Every Monday send me the funnel." "Remind me about Lina in 3 months."

Email it
From a team address. Orders from anyone else are ignored.

Assign it
Create a CRM task and assign it to the Buddy user, using the Tasks we already have.

Schedule it
Recurring jobs: briefing at 08:00, lead run at night, checks every 5 minutes.

The scheduler is a durable job queue in Buddy's own Postgres (pg-boss): retries, waits of months, no job runs twice, nothing is lost on restart. It never uses the CRM's Redis or worker.

How much freedom, per kind of action

Level	Buddy...	Examples
DO	acts and logs it	Read and search, answer questions, reports, notes, its own tasks, alerts to the team
DO AND TELL	acts, then tells the owner	Routine email replies once trusted, follow-up reminders, nurture emails to old contacts
ASK FIRST	drafts it and waits for a yes	A new kind of email reply, stage changes, bulk edits, merges, deletes, new workflows
NEVER	blocked in code	Change roles or settings, act beyond the user's role, decide an applicant's outcome, go over budget

Every kind of email reply starts as Ask first. It moves to Do and tell after two weeks with 90% or more of its drafts sent unchanged. You can see this score per kind of email at any time. This is how Buddy gets to replying on its own without surprising anyone.

What Buddy watches

What	How Buddy checks	Alert
A CRM workflow failed	Workflow runs with status FAILED, through the workflow tools and webhooks	Email to the owner within 5 minutes
A mailbox stopped syncing	Mailbox sync status through the API	Email, and in the daily report
CRM down or slow	Health endpoint and response times	Email immediately
Someone is interested	Every reply is sorted by intent (lead or applicant)	Email plus a CRM task for the owner
Buddy's own jobs	Run log, missed or doubled jobs	Email to the admin
Spending	Cost meter at 50, 80 and 100% of budget	Email; at 100% Buddy pauses
n8n and Apify runs	Their APIs	Email to the admin

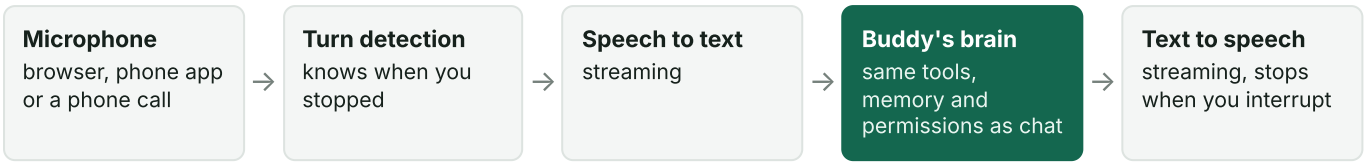
Daily report, 08:00, per person

New applicants and leads, replies waiting, people stuck in a stage, interested leads, tasks done and stuck, anything that failed, yesterday's cost and month to date. Each person's report only contains what their role can see. Weekly report on Fridays: funnel by stage and source, time in stage, form drop-off, what Buddy did.

CRM-SAFE Buddy only reads the CRM and adds notes, tasks and emails through it, as the person it acts for. It does not turn on, edit or delete existing CRM workflows. CRM workflows are still off in production, as FLOLABS.md says.

06 · VOICE

Talk to Buddy like a colleague



Why this shape. Anthropic does not offer a public real-time voice API (Claude's voice mode is only in its own apps), so Claude sits in the middle as text. That is good for us: voice is just a different way in and out of the same Buddy, with the same permissions and memory as chat.

Where to get each piece

Piece	Hosted (fastest to start)	Open source (self-hosted)
Framework	LiveKit Cloud	**Pipecat** (BSD-2, recommended) or LiveKit Agents (Apache-2.0; its turn model has its own license)
Speech to text	**Deepgram Nova-3**, about $0.005 to $0.008 a minute	faster-whisper or Whisper large-v3 (needs a GPU for real time); Kyutai STT (check license)
Text to speech	Cartesia Sonic or ElevenLabs Flash, about 3 cents per minute of speech	**Kokoro** (Apache-2.0, 82M, runs on CPU), Chatterbox (MIT), Orpheus (Apache-2.0)
Turn detection	Included in the hosted options	Silero VAD plus Pipecat's smart-turn model
Phone calls	Telnyx SIP, about $0.007 a minute plus about $1 a month per number	Works with both Pipecat and LiveKit (Telnyx has guides for each)
All in one		Kyutai Unmute: wraps any text model with open STT and TTS (check license before use)

1. Buddy app

A web app you can install on desktop and phone, with a push-to-talk button. No CRM change. First to ship.

2. Inside the CRM

A voice button as a CRM app. Our CRM already supports apps with a "Use your microphone" permission. Installed through the app system, not by editing the CRM. To confirm on our version in staging.

3. Phone

Call a number and talk to Buddy, through Telnyx. Later, once voice in the app works well.

Making it feel quick

- Target: first words back in about a second on office internet. Measured in phase 4, not promised up front.
- Streaming everywhere, a cached system prompt, and a "start speaking immediately" instruction to cut Claude's time to first word.
- Quick lookups go to Claude Haiku 5.5. Anything needing several tools goes to Sonnet 5.5, with a short "checking..." while it works.
- Language: Kokoro covers English and some other languages. If we need Arabic voice (Ahmed's forms), a hosted voice is the safer choice. To test.
- Today, without any work: the CRM's chatbot already has dictation (speech to text in the browser).

> CRM-SAFE The voice worker runs on Buddy's server and only talks to Buddy. Option 2 uses the CRM's own app system, and nothing is installed in production until it passes the staging tests.

New leads, old contacts, and a tap on the shoulder when someone bites

Target profile → Find (Apify) → Clean and dedupe → Enrich → Score → Add to CRM →
Reach out → Nurture old contacts → **Interest detected** → **Alert + task**

Step	How	Model and cost lever
Find	Apify actors (Google Maps places, website contact pages, company sites, job posts) through Apify's official MCP server, with our API key. Runs are asynchronous; results come back by webhook.	No AI
Clean and dedupe	Match against the CRM by email and domain, validate addresses, drop anyone already in the CRM or on the opt-out list.	Plain code
Enrich and score	Short website summary, size, location, fit. Rules first, then Claude on the shortlist.	Haiku 5.5, Batch API
Add to CRM	Only qualified leads, as new Company and Person records with a note "Source: Buddy lead search".	No AI
Reach out and nurture	Personal first email, two or three follow-ups, periodic check-ins with old contacts. Stops the moment someone replies.	Sonnet 5.5 (first), Haiku 5.5 (follow-ups), batched
Interest detected	Every reply sorted by intent. "Interested" means an email to the owner within 5 minutes and a CRM task.	Haiku 5.5
Daily lead report	Found, added, contacted, replied, interested, unsubscribed, bounced, cost.	Sonnet 5.5, batched

Guardrails for outreach

- A separate outreach domain and mailboxes, so cold email can never hurt hr@'s delivery to applicants.
- Warm-up, then a small daily cap per mailbox (start around 20), raised slowly.
- Unsubscribe link in every email, a shared opt-out list, quiet hours.
- All testing with internal +crmtest addresses first.

Legal, to check before the first send

- GDPR: a legitimate-interest check, and telling people where we got their details within a month (Art. 14).
- Cold email rules differ by country. Germany, for example, requires consent even for B2B.
- Site terms: no LinkedIn scraping, its terms forbid it.

Workflows

Buddy drafts CRM workflows with the CRM's own workflow tools (list, create steps, connect, activate). **Rule:** anything Buddy builds is created switched off and named "[Buddy] ...". A person reviews it and switches it on. For integrations outside the CRM (Calendly, Zoom, Slack), Buddy builds n8n workflows through n8n's API, on the n8n server we already run.

Apify cost: Starter plan about $29 a month, plus about $1 to $7 per 1,000 places, depending on the actor. Third-party figures, to confirm on apify.com.

> **CRM-SAFE** Leads are added as new records only. No existing record is changed without a person's yes. Raw scraped data stays in Buddy's database, and only qualified leads go into the CRM. New workflows are created switched off and existing ones are never edited.

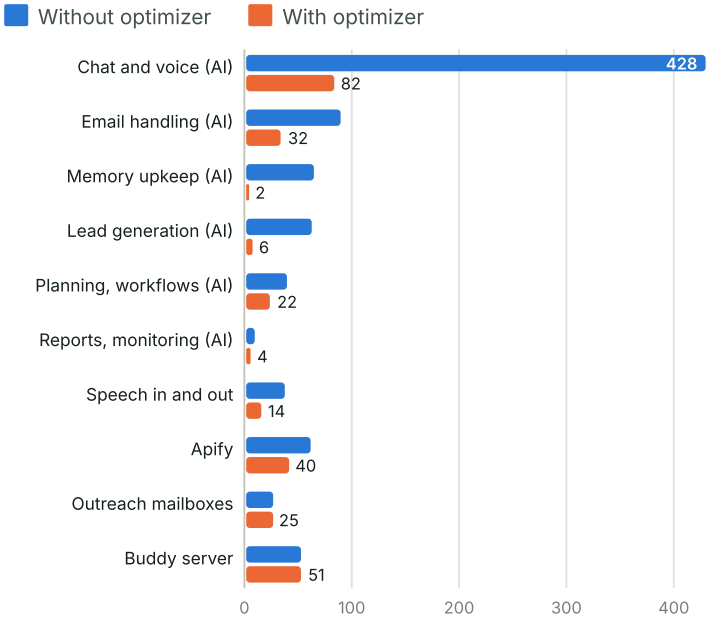
About $280 a month with the optimizer, about $860 without

An estimate for the usage below. The estimator in Buddy recalculates it from real usage after two weeks, so treat these as a starting point, not a quote.

Assumptions

People using Buddy	8, 22 working days
Chat and voice requests	15 / person / day
Model calls per request	about 4
Voice	10 min / person / day
Emails handled	150 a day, half replied
Leads found / first emails	3,000 / 600 a month
Old contacts nurtured	2,000 a month
Planning jobs on Opus 5.5	50 a month

Monthly cost by part (USD)



Claude prices used

Per million tokens	Input	Cache read	Output
Opus 5.5	$4	$0.20	$20
Sonnet 5.5	$2	$0.10	$10
Haiku 5.5 (prompt ≤ 100K)	$0.10	$0.01	$0.50

From Anthropic's pricing page, 31 Oct 2026. Cache writes cost 1.25x input. Batch API is 50% off everything. Haiku 5.5 costs 5x more above 100K-token prompts.

Part	Without	With optimizer	What changes
Chat and voice (AI)	$428	$82	Caching ($155), then half of the simple questions on Haiku 5.5
Email handling (AI)	$88	$32	Sorting on Haiku 5.5, cached template prompt for drafts
Memory upkeep (AI)	$63	$2	Haiku 5.5 through the Batch API at night
Lead generation (AI)	$61	$6	Batch API; Haiku for scoring and follow-ups
Planning, workflows (AI)	$38	$22	Caching; Opus 5.5 kept for hard jobs only
Reports, monitoring (AI)	$8	$4	Prepared through the Batch API before 08:00
Speech in and out	$36	$14	Local Kokoro voice instead of a hosted one
Apify	$60	$40	Capped results per run, cheaper actors, reuse
Outreach mailboxes	$25	$25	Domain plus two mailboxes
Buddy server	$51	$51	Same size as the CRM's (CPX32 with backups)
Total per month	**$858**	**$278**	**Embeddings: $0 with a local model (or about $3 to $9 once with Voyage)**

Not included: the CRM chatbot's own AI use (already metered in the CRM, and shown on Buddy's dashboard), phone calls (later), and one-off setup time. Speech, Apify and server prices are third-party or from FLOLABS.md, to confirm before buying.

09 · LIVE USAGE

See every token as it is spent

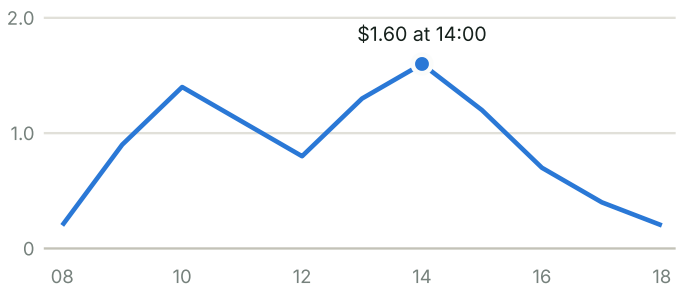
Cost is built in phase 1, before Buddy does any real work. Every call to Claude is metered the moment it happens, so we never find out about spending from the invoice.

Buddy · Cost and usage

Live · updated every few seconds · mock-up with example numbers

Spend today	Month to date	Cache hit rate	Tokens today
$9.84	**$212**	**87%**	**4.1M**
of a $15 daily cap	forecast $286 by 30 Nov	share of input read from cache	3.6M of them from cache

Spend per hour today (USD)



Monthly budget

$212 of $350 On track

61% used, 61% of the month gone. Alerts at 50, 80 and 100%; Buddy pauses at 100%.

Chat and voice	$5.10
Email	$2.20
Planning	$1.40
Leads, reports, memory	$1.14
CRM chatbot (from the CRM)	$3.10

Where the numbers come from

Source	What it gives	How often
Buddy's usage ledger	Every Claude call: model, input, cache write, cache read, output and its price, plus who asked, which feature and which task	Live
The CRM chatbot	Its token use and credits per turn, read from `agentTurn` through the API, read-only	Every minute
Anthropic usage and cost reports	Checks that our numbers match the bill. Buddy gets its own Anthropic workspace and key, so its spend is separate.	Daily
Speech, Apify, Telnyx	Their usage APIs, so the total covers more than AI tokens	Daily

What you can do from it

Drill down

By person, feature, model, task or day. See the most expensive tasks and what each one cost, down to a single call.

Set limits

Monthly budget, daily cap per person, cap per feature. Alerts by email at 50, 80 and 100%.

Stop it

Pause one feature (say, lead outreach) or all of Buddy with one switch. The CRM carries on as normal.

CRM-SAFE The dashboard lives in the Buddy app. A widget inside a CRM dashboard is optional, as a CRM app. The chatbot's usage is only read, never changed.

Spend fewer tokens without losing quality

The optimizer is a set of rules Buddy follows on every call, plus a weekly report that shows whether they are working. The estimator tells us what a change in usage will cost before we make it.

The optimizer: rules Buddy follows, and a weekly report on them

Lever	What it does	Effect in our estimate
Prompt caching	Tools and rules come first and stay identical, so after the first call they are re-read at 5% of the input price (Opus and Sonnet 5.5)	Chat $428 to $155
Model router	Haiku 5.5 for sorting, extracting and quick lookups (20x cheaper than Sonnet 5.5). Sonnet for conversation and drafts. Opus only for planning.	Chat $155 to $82; email $88 to $32
Batch API	50% off anything nobody waits for: lead scoring, memory upkeep, report preparation	Leads $61 to $6; memory $63 to $2
Plain queries first	Counts, filters and dedupe without AI. Claude only sees the shortlist, never 43k people.	Keeps cost flat as data grows
Budgets and kill switch	Daily cap per person, monthly cap per feature, hard stop for everything	Worst case is capped
Don't redo work	Reuse lead scores and summaries for 30 days. Keep Haiku prompts under 100K tokens.	Avoids repeats and the 5x tier
Weekly review	Top 10 costly tasks, cache hit rate, model mix. A change ships only if quality holds on a test set.	Keeps savings honest

The estimator

You enter

People using Buddy	8
Requests per person per day	15
Voice minutes per person per day	10
Emails handled per day	150
Leads found per month	3,000
Optimizer rules	on

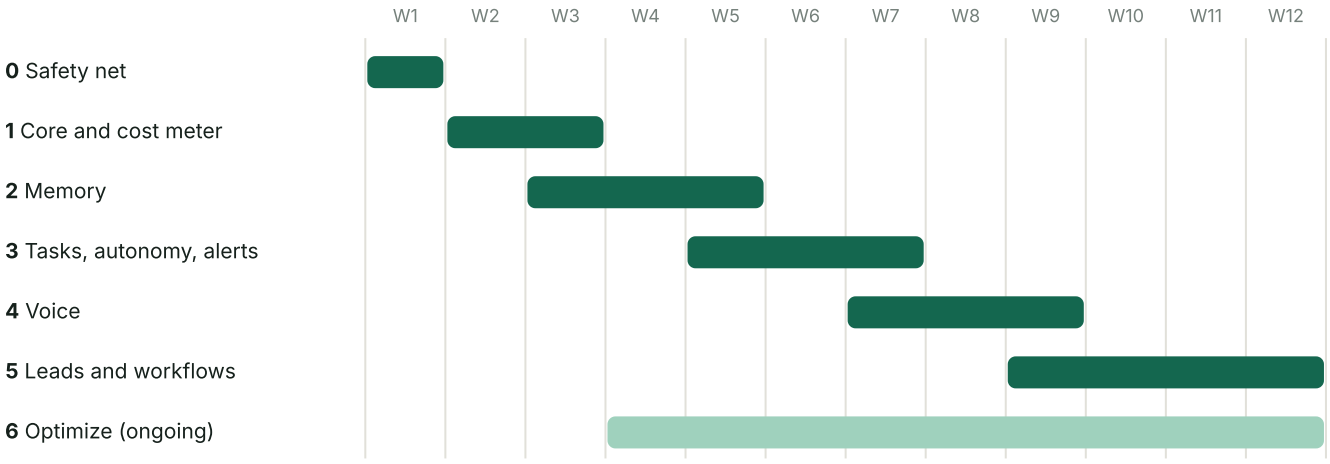
It shows

Monthly estimate	**$278**
AI tokens / speech / Apify	$148 / $14 / $40
Mailboxes and server	$76
Per person per month	about $35
Caching switched off	+$143
No model router (all Sonnet)	+$73

It uses the same price table as the live meter. For the first two weeks it works from the assumptions; after that it uses measured cost per completed task and forecasts the month. Prices are versioned, so a price change from Anthropic or a vendor shows up as a dated change, not a silent jump.

CRM-SAFE Optimizer and estimator run entirely inside Buddy. They change how Buddy calls Claude, never how the CRM works.

Seven phases, each useful on its own



Phase	Delivers	Done when (and tested how)
0. Safety net week 1	Staging server restored from a production backup. CRM regression suite (automated plus a short checklist) run twice as a baseline. Buddy user in the CRM. Buddy's OAuth app on staging. Accounts: Anthropic workspace and key, Apify, speech.	Regression suite passes twice in a row on staging. Login as a person works end to end.
1. Core and cost meter weeks 2 and 3	Buddy service, login as you, tools adapter with our tool list, text chat, usage ledger, live cost dashboard, budgets, kill switch, estimator.	A read-only user cannot change anything through Buddy (test). Buddy's numbers match Anthropic's report within 2%.
2. Memory weeks 3 to 5	Buddy Postgres, event log, webhook ingestion, slow night-time backfill, search by meaning, permission re-check, deletion sync, graph tables.	Right answers on a 50-question "who mentioned X" set. Nobody sees memory from records they can't open (test).
3. Tasks, autonomy, alerts weeks 5 to 7	Scheduler, tasks for Buddy, monitors and alerts, daily and weekly reports, email sorting and drafts, then Do and tell for trusted email types.	A week with no missed or doubled job. Alerts within 5 minutes. No auto-send until a type reaches 90% unchanged.
4. Voice weeks 7 to 9	Pipecat voice worker, speech in and out, voice in the Buddy app, optional CRM app button, speed tuning.	First words back in about a second on office internet. Same answers as chat on the test set.
5. Leads and workflows weeks 9 to 12	Apify through MCP, dedupe, scoring, outreach domain and warm-up, sequences, nurture, interest alerts, daily lead report, workflow drafts (switched off), n8n.	Two weeks with bounces under 2% and no complaints. Every "interested" reply alerted within 5 minutes.
6. Optimize from week 4	Weekly cost review, model routing checked against tests, graph memory upgrade if search needs it.	Cost per completed task falls or holds while quality holds.

CRM-SAFE CRM code changes in every phase: **none**. Each phase ends with the CRM regression suite on staging, before and after Buddy is switched on, and goes to production only after both pass.

How we prove nothing broke, every time

1. Fresh copy Restore the latest production backup to staging → **2. Test the CRM** Full regression suite, Buddy off → **3. Switch Buddy on** Run Buddy's own tests on staging → **4. Test the CRM again** Same suite, plus speed and error rates → **5. Production** Only if both runs pass, right after a backup

CRM regression suite covers	How
Login and permissions	Admin, editor and read-only users each try what they should and shouldn't be able to do
People, Applications, views and filters	Open, filter, edit and save, on production-sized data
Forms	Submit a public form, check the person, the answers and the timeline
Team inbox and email	Sync, read state, send and reply from a +crmtest address
Workflows and the chatbot	Workflows page loads, a test run works, the chatbot answers and uses tools
Speed	Page and API response times before and after Buddy, within 10%

Automated with Playwright (the repo already has a `twenty-e2e-testing` package), plus a 15-minute manual checklist. **Rollback:** stop Buddy. The CRM keeps working because nothing in it changed, and every record Buddy made can be listed from its log and reverted.

Why restore staging from a backup: on 30 Oct we found that a fresh, empty CRM install currently fails to create its first workspace (the FORM_SUBMISSIONS widget type issue). Production is not affected. Restoring a backup avoids the problem and tests on real-sized data.

Risks and the plan for each

Risk	Plan
Buddy shows someone data they can't see	Acts with their role. Memory is re-checked against the CRM with their login before it is shown.
Buddy slows the CRM	Own server. The CRM's own rate limits for apps. Night-time, throttled backfills. Speed is part of the regression suite.
A wrong or unwanted email goes out	Drafts first; trust is earned per email type. Daily caps, quiet hours, +crmtest addresses in testing.
Instructions hidden in an email from an applicant or lead	Treated as data, never as orders. Only team members can instruct Buddy. Outgoing actions follow the autonomy levels whatever the email says.
Costs run away	Live meter, caps per person and feature, alerts at 50, 80 and 100%, pause at 100%.
Cold outreach hurts applicant email	Separate domain and mailboxes, warm-up, low volume, unsubscribe and opt-out list.
Legal: GDPR, cold email, AI in hiring	Legitimate-interest check and source notice for leads; country rules; no automated hiring decisions; a DPIA before anything ranks applicants.
A CRM upgrade breaks Buddy	Contract tests on every upgrade. Only the tools adapter knows the CRM, so a fix lands in one place.

CRM-SAFE The test steps above run for every Buddy release. A Buddy release never ships together with a CRM release, so if something changes we know which one did it.

What we need from you, and what we build on

1 The two rules
Approve them, and API-first (option A), for every phase.

2 Where it runs
Own server (recommended, ~$51/month) or a capped container.

3 Voice
Hosted (Deepgram, Cartesia) or local (Kokoro). Arabic needed?

4 Embeddings
Local model (recommended) or Voyage.

5 Email autonomy
Which email types may send without approval after the trial.

6 Lead generation
Target profile, countries, outreach domain, daily caps.

7 Budget
Monthly AI budget and daily cap per person.

8 Roles
Who sets the Buddy user's role and the app ceiling.

Open-source projects we can use

Need	Repository	License	Use
Voice	`pipecat-ai/pipecat`	BSD-2	Voice worker (recommended)
Voice	`livekit/agents`	Apache-2.0*	Alternative, rooms and telephony at scale
Voice	`kyutai-labs/unmute`	check	Try for a fully local voice
Speech	`hexgrad/kokoro`, `resemble-ai/chatterbox`	Apache-2.0, MIT	Local text to speech
Speech	`SYSTRAN/faster-whisper`	MIT	Local speech to text (GPU for real time)
Memory	`pgvector/pgvector`, `apache/age`	PostgreSQL, Apache-2.0	Search by meaning, graph in Postgres
Memory	`getzep/graphiti`, `mem0ai/mem0`	Apache-2.0	Temporal graph (trial), preferences memory
Jobs	`timgit/pg-boss`	MIT	Scheduler on Buddy's Postgres
Leads	`apify/apify-mcp-server`	MIT	Apify actors as tools
Leads	`Kandid-ai/AgentSDR`	MIT	Reference for sequences and reply sorting
Cost	`langfuse/langfuse`	MIT core	Traces and charts, only if we outgrow our own meter

* LiveKit's turn-detection model has its own license that limits it to LiveKit Agents. Licenses were taken from the sources below; check each repository before use.

Sources

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